# 互联网协议实验

- 搭建实验环境
  - **$** sudo apt install wireshark ifupdown
  - $ sudo mn --nat  # allows hosts to connect to the Internet
  - **mininet>** xterm h1
  - **h1 #** echo "nameserver 8.8.8.8" > /etc/resolv.conf
  - h1 # wireshark &

- 实验步骤
  - h1 # wget -U "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/123.0 Safari/537.36" https://www.ucas.ac.cn

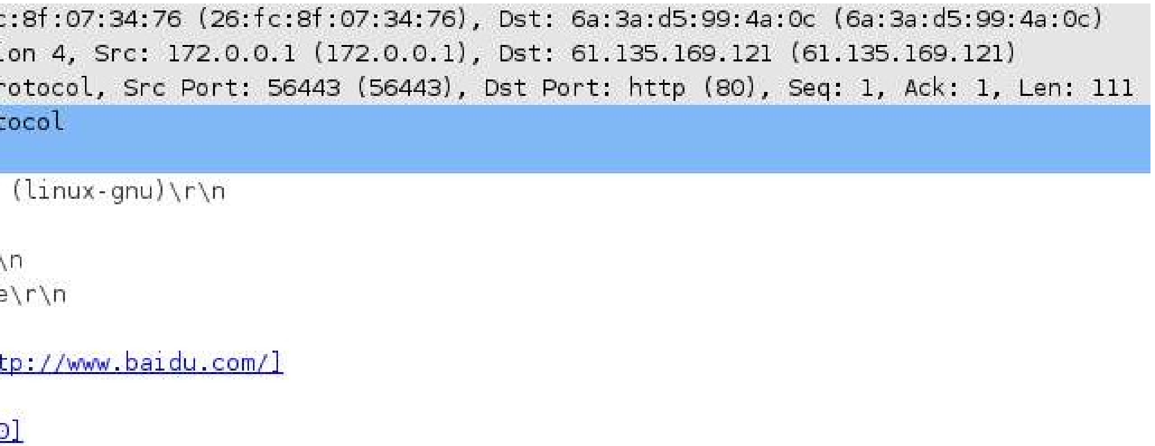
# 互联网协议实验

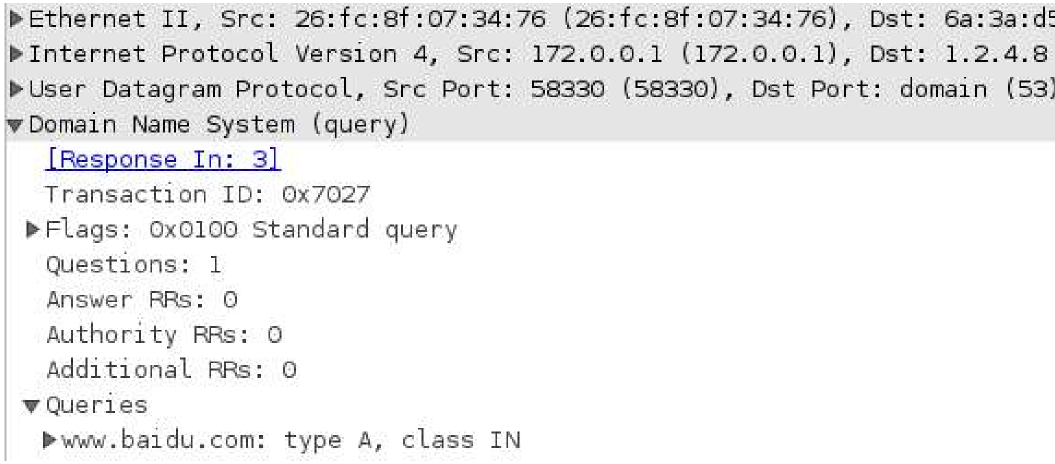
- 观察wireshark输出 （一）， 以www.baidu.com页面为例
  - ARP协议: IP地址->MAC地址映射
  - DNS协议: 域名->IP地址映射
  - TCP协议: 数据传输

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 4 | 9.256069038 | 82:24:2a:96:c2:c8 | Broadcast | ARP | 42 | Who has 10.0.0.3? Tell 10.0.0.1 |
| 5 | 9.259799814 | 72:e0:11:58:ce:4f | 82:24:2a:96:c2:c8 | ARP | 42 | 10.0.0.3 is at 72:e0:11:58:ce:4f |
| 6 | 9.259812291 | 10.0.0.1 | 159.226.39.1 | DNS | 73 | Standard query 0x0393 A www.baidu.com |
| 7 | 9.259815360 | 10.0.0.1 | 159.226.39.1 | DNS | 73 | Standard query 0xd6fe AAAA www.baidu.com |
| 8 | 9.308805004 | 159.226.39.1 | 10.0.0.1 | DNS | 132 | Standard query response 0x0393 A www.baidu.com CNAME w |
| 9 | 9.318736091 | 159.226.39.1 | 10.0.0.1 | DNS | 157 | Standard query response 0xd6fe AAAA www.baidu.com CNAM |
| 10 | 9.319227342 | 10.0.0.1 | 119.75.216.20 | TCP | 74 | 48488 → 80 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_P |
| 11 | 9.338055235 | 119.75.216.20 | 10.0.0.1 | TCP | 58 | 80 → 48488 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS= |
| 12 | 9.338087433 | 10.0.0.1 | 119.75.216.20 | TCP | 54 | 48488 → 80 [ACK] Seq=1 Ack=1 Win=29200 Len=0 |
| 13 | 9.341933425 | 10.0.0.1 | 119.75.216.20 | HTTP | 194 | GET / HTTP/1.1 |
| 14 | 9.342532422 | 119.75.216.20 | 10.0.0.1 | TCP | 54 | 80 → 48488 [ACK] Seq=1 Ack=141 Win=65535 Len=0 |
| 15 | 9.353627861 | 119.75.216.20 | 10.0.0.1 | TCP | 454 | [TCP segment of a reassembled PDU] |
| 16 | 9.353647250 | 10.0.0.1 | 119.75.216.20 | TCP | 54 | 48488 → 80 [ACK] Seq=141 Ack=401 Win=30016 Len=0 |
| 17 | 9.362959249 | 119.75.216.20 | 10.0.0.1 | TCP | 1474 | [TCP segment of a reassembled PDU] |
| 18 | 9.362983464 | 10.0.0.1 | 119.75.216.20 | TCP | 54 | 48488 → 80 [ACK] Seq=141 Ack=1821 Win=32660 Len=0 |
| 19 | 9.363141969 | 119.75.216.20 | 10.0.0.1 | HTTP | 1015 | HTTP/1.1 200 OK (text/html) |
| 20 | 9.363161257 | 10.0.0.1 | 119.75.216.20 | TCP | 54 | 48488 → 80 [ACK] Seq=141 Ack=2782 Win=35500 Len=0 |
| 21 | 9.363337625 | 119.75.216.20 | 10.0.0.1 | TCP | 54 | 80 → 48488 [FIN, ACK] Seq=2782 Ack=141 Win=65535 Len=0 |
| 22 | 9.366398955 | 10.0.0.1 | 119.75.216.20 | TCP | 54 | 48488 → 80 [FIN, ACK] Seq=141 Ack=2783 Win=35500 Len=0 |
| 23 | 9.367574703 | 119.75.216.20 | 10.0.0.1 | TCP | 54 | 80 → 48488 [ACK] Seq=2783 Ack=142 Win=65535 Len=0 |

# 互联网协议实验

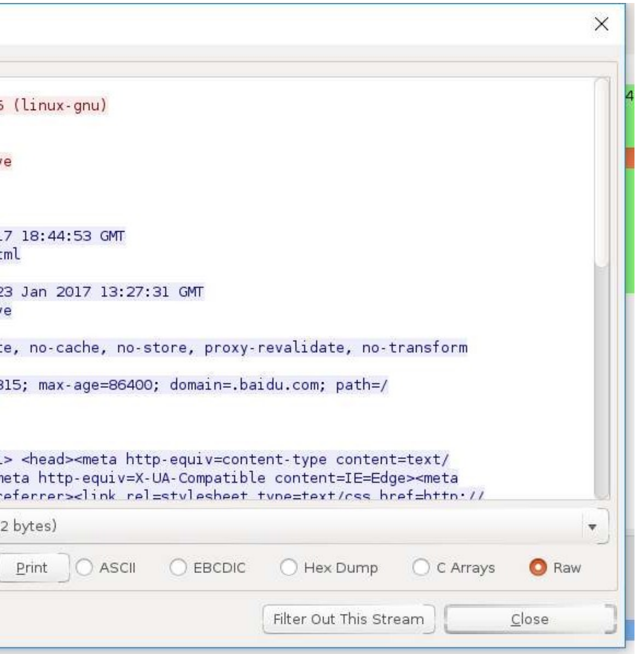
- 观察Wireshark输出 （二）
  - 不同层次的协议封装
- Ethernet < IP < UDP < DNS
- Ethernet < IP < TCP < HTTP

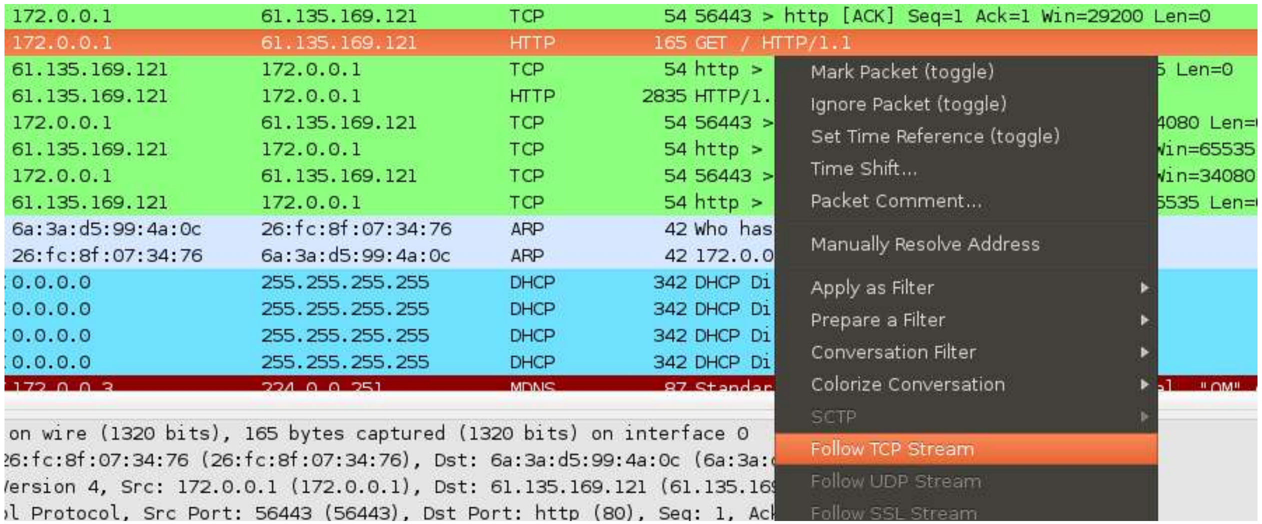


# 互联网协议实验

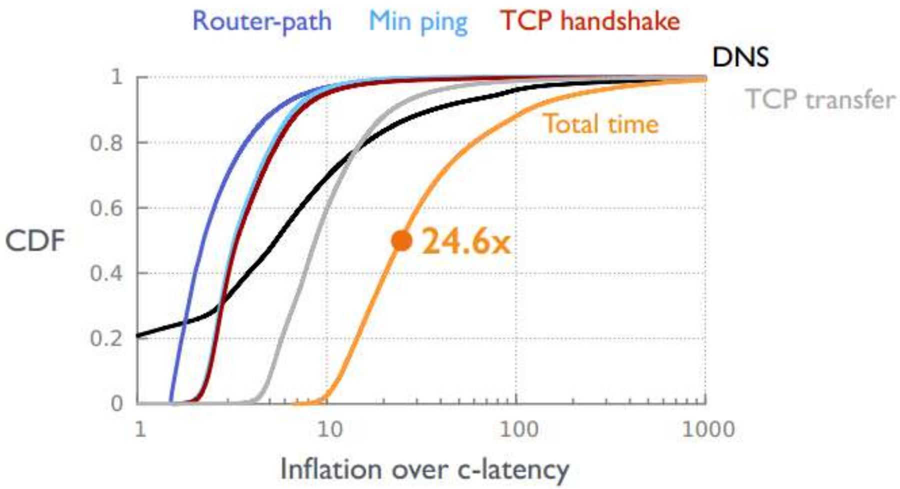
- 观察Wireshark输出（三）
  - TCP承载HTTP协议



# 传输性能测量实验

- 在Linux系统或Mininet环境下，编写Python测量脚本，复现如下实验结果
  - 只需要Ping、DNS、TCP transfer、Total time这几个值
  - 获取1000个左右网站的上述指标
  - 使用GeoIP数据库查询IP地址对应的地理位置，大致估算c-latency



# 实验内容

- 互联网协议实验
  - 在节点h1上开启wireshark抓包，用wget下载**www.ucas.ac.cn**页面
  - 调研说明wireshark抓到的几种协议
    - ARP, DNS, TCP, HTTP, HTTPS
  - 调研解释h1下载ucas页面的整个过程
    - 几种协议的运行机制
- 传输性能测量实验
  - 如果在Mininet环境下测量，$ sudo mn **--nat**
  - 测量结果可能与原图不太一样：当前大部分通过CDN技术分发，服务器地址大都在北京
  - 结合本讲的内容，调研解释这幅图中结果的成因